SortSense

AI-Based Waste Identification & Segregation — Project Report

Sociothon 2026 · NSS IIT Roorkee · Smart and Sustainable Waste Management

Team: **team_name** — Arpit Seth, Rajeev Maheshwari, Garv Agarwal

"Every item in the right bin — decided in under a second, at the point of disposal."

1. Problem Addressed

Segregation fails at the very first second — when a person stands in front of a bin holding an item. Biodegradable, recyclable, non-recyclable and hazardous waste are mixed because people are unsure where items go (chip packets, paper cups, tetra packs, batteries), bins give no feedback, and no one measures segregation quality. Mixed waste contaminates recyclables, forces sorting staff to hand-pick through dirty waste, and sends recoverable material to landfill. Institutions also lack data to target improvement.

SortSense addresses **Area A — AI-Based Waste Identification and Segregation** of the Sociothon problem statement, and feeds data into Area D (monitoring).

2. Solution Explanation

2.1 Proposed approach

SortSense is a low-cost **retrofit segregation station** mounted on existing 3/4-bin sets, plus a companion **web/mobile app**. A camera captures the item, an on-device convolutional neural network classifies it into one of four categories, and the station guides the user (LED, voice, auto-opening lid) to the correct compartment. Every decision is logged to a dashboard.

Category	Examples	Destination
Biodegradable	Food waste, peels, tea leaves, leaves	Composter / biogas
Recyclable	PET bottles, cans, paper, cardboard, glass	Authorised recycler / kabadiwala
Non-recyclable	Multi-layer wrappers, soiled tissue, thermocol	RDF / landfill
Hazardous / E-waste	Batteries, bulbs, chargers, sanitary/medical	Authorised hazardous handler

2.2 How it works

Capture — ESP32-CAM at the bin (IR-triggered) or the phone app. **Classify** — MobileNetV2 fine-tuned on TrashNet, TACO and campus images (YOLOv8n for multiple items). **Decide** — if confidence ≥ 80% the item is auto-routed; below that the station asks the user to press one of four buttons. **Route** — a servo-driven chute/lid opens the right compartment. **Learn** — each human confirmation is stored as a labelled image for monthly retraining; fill sensors report bin levels.

2.3 Handling classification errors

The problem statement asks how errors are handled. SortSense never forces a guess: (1) a confidence threshold routes uncertain items to a human, (2) hazardous predictions are always treated

conservatively (any hazardous signal → red bin + alert), (3) contamination detection flags food residue headed to the recyclable bin, (4) the human-in-the-loop labels continuously reduce error on local items. The training script reports overall accuracy, accuracy on auto-sorted items and the share sent to human confirmation, so the threshold can be tuned with data.

2.4 Target users

Students and staff disposing waste; housekeeping and sorting teams; the estate office / sanitation committee; NSS volunteers running awareness drives; later, RWAs and municipal wards.

2.5 Working prototype (submitted)

prototype/index.html — a browser app running MobileNetV2 fully on-device via TensorFlow.js. It accepts an uploaded photo, live camera capture or a typed item name; maps the prediction to the four categories with disposal instructions; applies the 80% confidence rule with a human-confirm fallback; animates a 4-compartment smart bin; and shows a live dashboard (items sorted, auto-sort rate, recyclables recovered, category chart, CSV export). **model/train_sortsense.py** trains the custom 4-class model and exports an int8 TFLite file; **hardware/sortsense_bin.ino** is the ESP32-CAM station firmware.

3. Impact Assessment

All figures below are transparent estimates to be validated in the pilot, as the guidelines recommend.

Assumption	Value
Campus residents (students + staff)	8,000
Waste per person per day	0.3 kg → 2.4 t/day
Recyclable share of waste	25% → 600 kg/day
Clean recovery of recyclables today (mixed bins)	~40% → 240 kg/day
Clean recovery with SortSense (target)	~75% → 450 kg/day
Additional recyclables recovered	≈ 210 kg/day ≈ 75 t/year

Metric	How we measure it	Target
Classification accuracy	Held-out test set (train script metrics.txt)	≥ 85%
Accuracy on auto-sorted items	Items with conf ≥ 0.80	≥ 95%
Segregation rate at pilot bins	Weekly audit of 50 random items/bin vs. baseline week	+30 pp
Landfill diversion	Weighed recyclable + compost output	+75 t/yr per campus
Sorting labour	Staff minutes spent re-sorting per day	–40%
Decision latency	On-device inference time	< 1 s

Social impact: cleaner hostels and canteens; safer work for sanitation staff (less hand-sorting, hazardous items isolated); better-quality material for informal recyclers, raising its value; and data for the administration to target awareness drives at the hostels that need them most.

4. Implementation Plan

4.1 Resources and estimated cost

Item	Qty (pilot)	Unit (Rs.)	Total (Rs.)
ESP32-CAM module	10	700	7,000
Servo chute + mount	10	600	6,000
Fill sensors ×3, LED, buzzer, buttons	10	450	4,500
Power supply & wiring	10	650	6,500
Enclosure / retrofit on existing bins	10	1,600	16,000
Raspberry Pi 4 gateway (per building)	2	6,500	13,000
Misc. & spares	—	—	2,000
Pilot total (10 stations, 2 buildings)			≈ 55,000

Software uses only open-source tools (TensorFlow, FastAPI, SQLite/PostgreSQL, Chart.js). The phone/web app has zero hardware cost and can launch campus-wide on day one. Running cost: electricity (<5 W per station) and one volunteer-hour per week for label review.

4.2 Deployment process

Phase	Timeline	Activities
0 · Data	Week 1–2	Collect ~3,000 campus photos via app; combine with TrashNet/TACO; label into 4 classes
1 · Pilot	Week 3–6	10 stations in one hostel + canteen; baseline audit week then SortSense weeks; tune threshold
2 · Campus	Month 2–3	All hostels and canteens; dashboard handed to estate office; hostel leaderboard with NSS
3 · Scale	Month 4+	Partner campuses and RWAs; app-only mode for low-budget sites; municipal ward pilots

4.3 Potential challenges and mitigation

Challenge	Mitigation
Poor lighting, dirty or crushed items	Augmented training data, built-in LED light, human-confirm fallback
Local items not in public datasets	Human-in-the-loop labelling; monthly retraining
Vandalism / weather	Sealed enclosure, indoor placement first
User fatigue	Sub-second response, leaderboards, NSS awareness drives
Privacy	Camera points only at the tray; no faces stored; images used only for training

4.4 Scaling plan

One hostel → all campus hostels and canteens → other institutions (same model, retrained with their images) → municipal wards and RWAs. The app-only version scales at near-zero cost; hardware stations scale linearly at ≈ Rs. 4,000 each, far below commercial smart bins, because they retrofit

existing bins instead of replacing them. Dashboards from multiple sites can be compared to share best practices.

5. Conclusion

SortSense makes the first second of disposal smart, cheap and measurable: it identifies the item, routes it correctly, admits when it is unsure, and turns every disposal into data that improves both the model and the campus’s segregation habits.